

16S rRNA vs Shotgun Metagenomics

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Summary

Why Study Microbial Communities?

The Challenge

- **~99% of microbes** cannot be cultured in the lab
- Communities: 10^3 – 10^9 species per gram
- Single-isolate sequencing misses most diversity
- Function emerges from community interactions

Applications

- Human gut health & disease biomarkers
- Soil & environmental ecology
- Ocean carbon cycling
- Clinical infection profiling
- Fermentation & food science

Two Culture-Independent Approaches

16S rRNA Amplicon

Targets universal bacterial/archaeal gene marker

Shotgun Metagenomics

Sequences ALL DNA in the sample — no targeting needed

16S rRNA Amplicon Sequencing

The 16S ribosomal RNA gene (~1,550 bp) contains 9 hypervariable regions (V1–V9) flanked by conserved sequences — making it ideal for universal PCR amplification and species-level discrimination. V3–V4 and V4 are the most commonly used regions.

1

DNA Extraction

From soil, feces, water... Remove inhibitors

PowerSoil Kit, Qiagen

2

PCR Amplification

Amplify target hypervariable region (V3-V4 most common)

Universal primers 515F/806R

3

Library Prep & Sequencing

Paired-end Illumina (MiSeq) or PacBio for full-length

Illumina MiSeq, PacBio

4

Bioinformatics (DADA2)

Denosing → ASVs; OTU clustering at 97% similarity

DADA2, QIIME2

5

Diversity & Taxonomy

Assign taxonomy, alpha/beta diversity, stats

SILVA, phyloseq, vegan

QIIME2 Analysis Pipeline — Step by Step

Import & Demultiplex

```
qiime tools import  
qiime demux emp-paired
```

Convert raw FASTQ; separate samples by barcode

Quality Filter & Trim

```
qiime cutadapt trim-paired
```

Visualise quality; trim primers

Denoise (DADA2)

```
qiime dada2 denoise-paired
```

Error model learning; ASV feature table

Phylogenetic Tree

```
qiime phylogeny align-to-tree-  
mafft-fasttree
```

MAFFT alignment → FastTree → rooted tree

Taxonomy Classification

```
qiime feature-classifier  
classify-sklearn
```

Naive Bayes classifier trained on SILVA/Greengenes

Diversity Analysis

```
qiime diversity  
core-metrics-phylogenetic
```

Alpha (Shannon, Faith PD) & beta (UniFrac, Bray-Curtis)

Differential Abundance

ANCOM / DESeq2 / ALDEx2

Identify taxa significantly differing between groups

Output: Feature table (samples × ASVs) · Taxonomy bar charts · PCoA plots · Alpha diversity box plots · Differential abundance

Shotgun Metagenomics

Shotgun metagenomics fragments ALL DNA in a sample and sequences it without any amplification step. This gives access to bacteria, archaea, viruses, fungi, and host DNA — plus functional genes and metabolic pathways.

1

DNA Extraction

High-quality, high-MW DNA. Remove inhibitors.

2

Library Preparation

Fragment, ligate adapters, optional size selection

3

Sequencing

Short reads (NovaSeq) or long reads (Nanopore/PacBio)

4

QC & Host Removal

Trim adapters, filter low quality, remove host reads

5

Taxonomic Profiling

Kraken2/Bracken or MetaPhlAn4 — species-level IDs

6

Assembly & Binning

MEGAHIT → MetaBAT2 → MAGs

7

Functional Annotation

HUMAN3, eggNOG-mapper → pathways & ARGs

Functional Outputs (unique to shotgun)

- Metabolic pathway abundance (KEGG, MetaCyc)
- Antibiotic resistance genes (ARGs)
- Virulence factors
- Metagenome-Assembled Genomes (MAGs)
- Novel gene discovery

Cost consideration

5–10× more expensive than 16S. Requires 10–30 Gb/sample (vs ~50,000 reads for 16S). Significant computational resources needed.

Kingdoms detected

Bacteria Archaea Fungi Viruses Parasites Host DNA

16S vs Shotgun

Feature	16S rRNA Amplicon	Shotgun Metagenomics
Target	16S rRNA gene only	Entire metagenome (all DNA)
Kingdoms	Bacteria & Archaea only	All kingdoms: Bacteria, Archaea, Fungi, Viruses
Taxonomic resolution	Genus level (species w/ ASVs)	Species to strain level
Functional data	None	Full pathways, ARGs, virulence factors
PCR bias	Yes — amplification bias	No PCR step (unbiased)
Cost/sample	~\$30–80	~\$200–600
Sequencing depth	50,000–100,000 reads	10–30 Gb per sample
Computational load	Moderate	High — requires HPC or cloud
Standard pipeline	QIIME2 + DADA2	Kraken2/MetaPhlAn4 + HUMAnN3
Best for	Community surveys, large cohorts	Functional studies, novel organisms, ARGs

When to Use Which Method?

Choose 16S rRNA when...

- Quick, cost-effective community overview needed
- Large sample size (dozens–hundreds of samples)
- Budget is limited (~\$30–80/sample)
- Only Bacteria/Archaea are relevant
- Comparing against large existing databases (HMP)
- Research question: "Who is there?" not "What do they do?"
- Clinical screening or large patient cohort studies

Choose Shotgun when...

- Functional gene/pathway information needed
- Viruses, fungi, or eukaryotes are relevant
- Strain-level or subspecies resolution required
- Searching for antibiotic resistance genes (ARGs)
- Recovering novel genomes (MAGs) is a goal
- Budget and HPC resources are available
- Research question: "What are they doing?"

Pro tip: Many large studies use BOTH — 16S for broad community profiling across the full cohort, then shotgun on a subset for deep functional validation.

Diversity Metrics & Statistical Analysis

Alpha Diversity — within-sample

Observed features

Number of distinct ASVs/species

Shannon index

Richness + evenness combined

Faith's PD

Phylogenetic diversity (16S)

Chao1

Estimates true richness incl. rare taxa

Beta Diversity — between-sample

Bray-Curtis — abundance-weighted; most common

Weighted UniFrac — phylogenetic + abundance

Unweighted UniFrac — presence/absence + phylogenetic

Jaccard — presence/absence only

Note: Always rarefy reads before alpha diversity comparisons to normalize sequencing depth

Statistical Tests

PERMANOVA (adonis2) — group differences in beta diversity

Kruskal-Wallis — non-parametric per-taxon test

ANCOM / DESeq2 / ALDEx2 — differential abundance

SpiecEasi / CoNet — co-occurrence networks

Key R Packages

phyloseq	Data management & visualisation
vegan	Community ecology statistics
microbiome	Utilities built on phyloseq
ggplot2	Publication-quality plots

Shotgun Analysis Tools — Read-based & Assembly

Read-Based Profiling

Classify reads directly — fast, no assembly needed

Kraken2 + Bracken

K-mer matching against a reference DB. Extremely fast. Bracken for abundance re-estimation at species level.

MetaPhlAn 4

Marker-gene based profiling. Species-level with strain tracking. Standard in large-scale studies (HMP, curatedMetagenomicData).

HUMAnN 3

Pathway/gene family abundances from MetaPhlAn + UniRef. Gives complete functional profile.

StrainPhlAn

Strain-level tracking using MetaPhlAn marker SNPs. Useful for longitudinal & transmission studies.

Assembly-Based (MAGs)

Assemble reads into genomes — recover novel organisms

MEGAHIT / SPAdes

De novo assembly of contigs from short reads. MEGAHIT is faster for large datasets; SPAdes gives higher contiguity.

MetaBAT2 / CONCOCT

Bin contigs into draft genomes (MAGs) using coverage + tetranucleotide frequency signatures.

CheckM

Assess MAG completeness and contamination using a curated set of lineage-specific marker genes.

GTDB-Tk + Prokka

Taxonomic classification of MAGs against GTDB; Prokka/eggNOG-mapper for functional annotation.

Tip: Most shotgun studies use both tracks — profiling for rapid community overview, assembly for novel genome recovery and deep functional analysis.

Summary & Take-Home Messages

1

16S is targeted

Amplifies the universal bacterial 16S marker gene. Cheaper, simpler, but limited to Bacteria + Archaea with no functional data.

2

Shotgun is unbiased

Sequences ALL DNA. Detects all kingdoms, provides functional gene profiles, enables MAG recovery. More powerful but more expensive.

3

ASVs over OTUs

DADA2 ASVs are now preferred over 97% OTU clustering — more reproducible, higher resolution, no arbitrary cutoff.

4

Standard pipelines

16S → QIIME2 + DADA2. Shotgun → KneadData → Kraken2/MetaPhlAn4 → HUMAnN3 or MEGAHIT → MetaBAT2.

5

Always report diversity

Alpha (Shannon, Faith PD, observed) AND beta diversity (Bray-Curtis, UniFrac) with appropriate statistics (PERMANOVA).

6

Match method to question

Community survey on a budget → 16S. Functional genes, viruses, novel organisms → Shotgun. Large study → both.